Machine-Verifiable CDU Safety Procedures: A Worked Extraction from OCP Project Deschutes

Nauman Rafique, Co-founder, Visum AI

Marco Pritoni, Research Scientist, Lawrence Berkeley National Laboratory

Other contributors: Leonardo Bachega, Co-founder, Visum AI

An Alarm Fires at 3am



- The coolant distribution unit reports coolant in one of three rope sensor zones.
- The technician on shift opens the leak response EOP. It was written against the unit as it was installed, and the unit has changed since.

What a Procedure Mistake Costs

59%

of human error outages: failure to follow procedures

36%

incorrect staff processes and procedures

87%

could have been avoided with better process or configuration

1 in 5

of major outages cost more than $1 million, two years running

- That work happens under time pressure. When it goes wrong, it lands in these numbers: the procedure was not followed, or the procedure itself was wrong.*

* Uptime Institute, 2026 Annual Outage Analysis.

Procedures Are Hard to Keep Correct

1 MW

where AI factory racks are heading, liquid cooled, moving to high-voltage DC

Google, OCP EMEA Summit 2025

800+

procedures running in a modern data center

Days

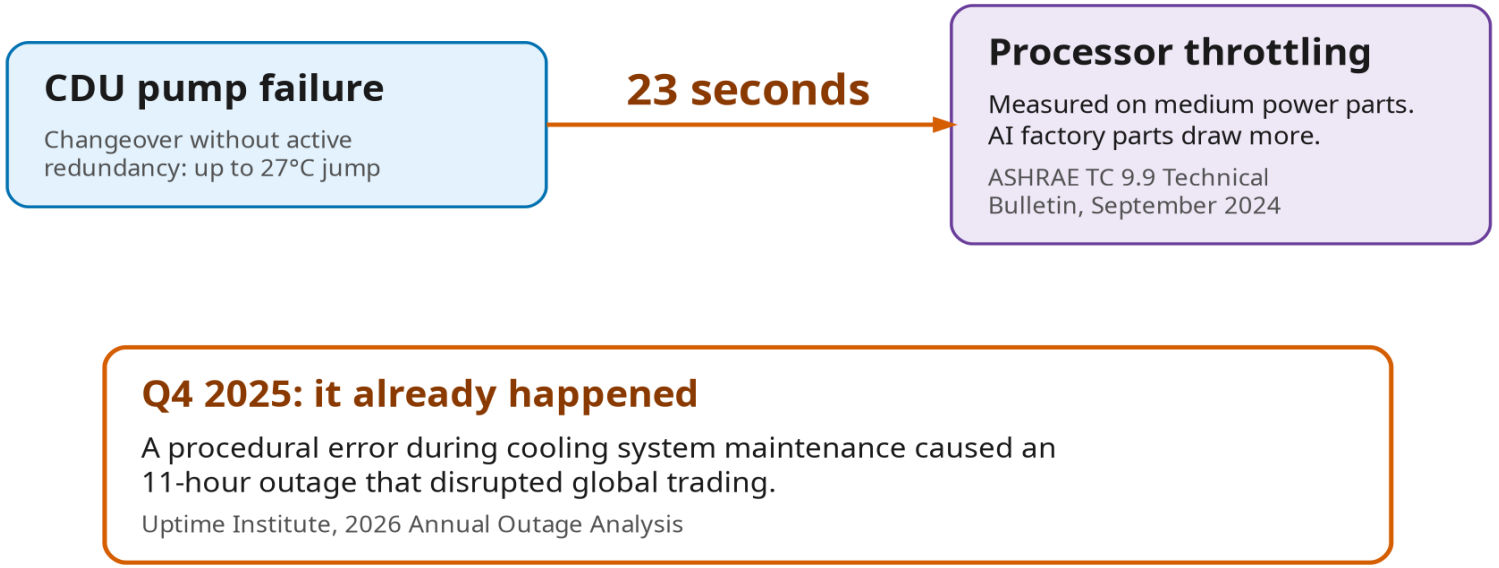
for a senior engineer to write one procedure by hand

- Roughly two thirds of FY2025 lockout-tagout citations concerned the written procedure, its training, and its inspection, not the hardware.*
- 25% of operators report staff hired away by competitors, against 12% who lose them outside the industry. The expertise leaves the building, not the field.†
- One procedure draws on the vendor manual, the drawings, the commissioning record, and the safety standard. When any of those moves, the procedure goes back on a senior engineer's desk for days.

* OSHA FY2025 Top 10.

† Uptime Institute Global Data Center Survey 2025.

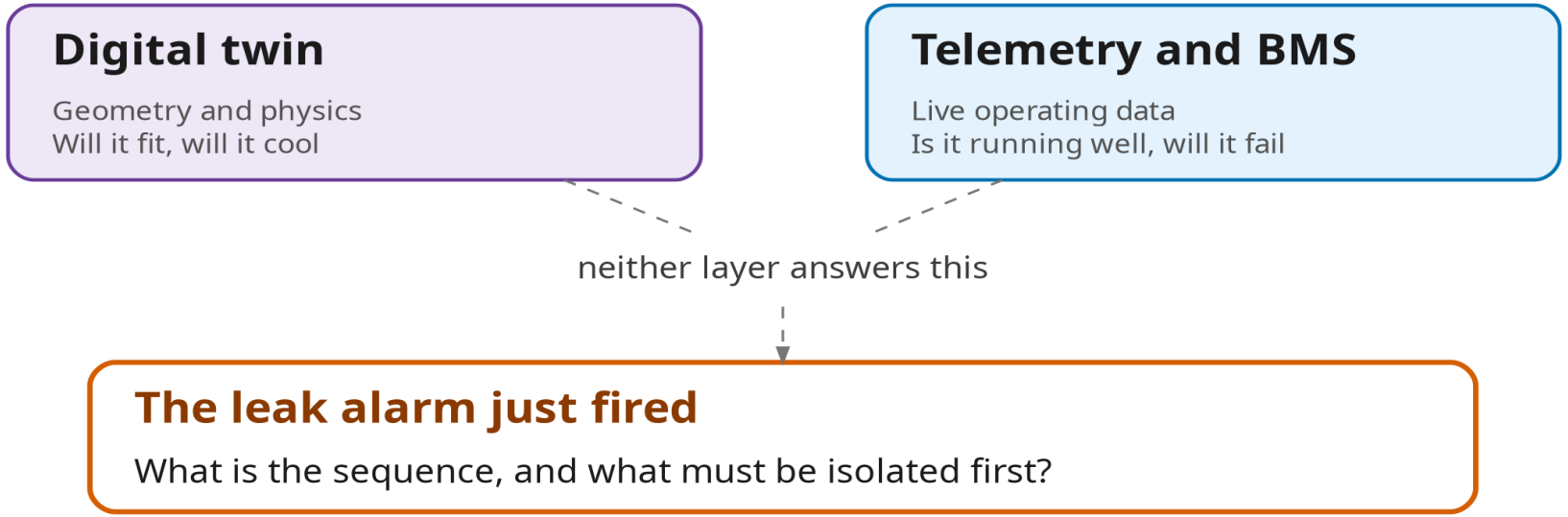
Liquid Cooling and 800 V DC Raise the Stakes



- Liquid cooling adds isolation points, interlocks, and failure modes to every procedure that touches it. The move to 800 V DC distribution adds more.
- ASHRAE's May 2026 bulletin reports safeties found left in an override position after commissioning, with the MOP and the SOP named as the evidence of record.*

* ASHRAE TC 9.9 Technical Bulletins, September 2024 and May 2026.

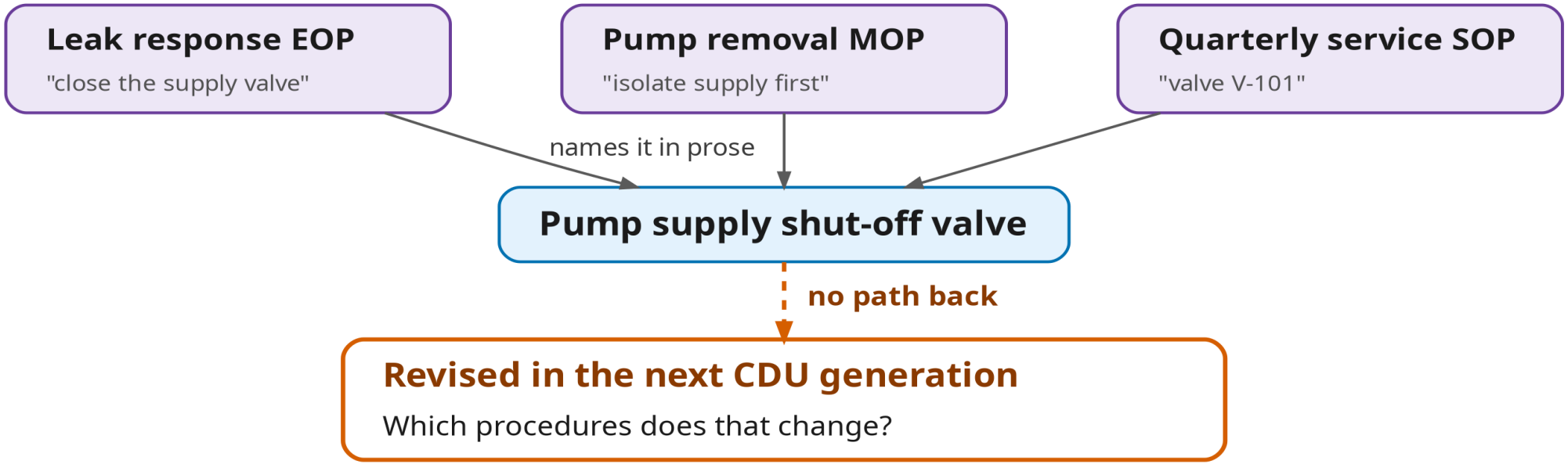
The Facility Is Modeled. The Work Is Not.



- Brick and ASHRAE 223 already model the equipment and its telemetry.
- Nothing models the work itself: the steps, the isolation points, the interlocks, the order they happen in. So every tool that needs the work invents its own representation.*

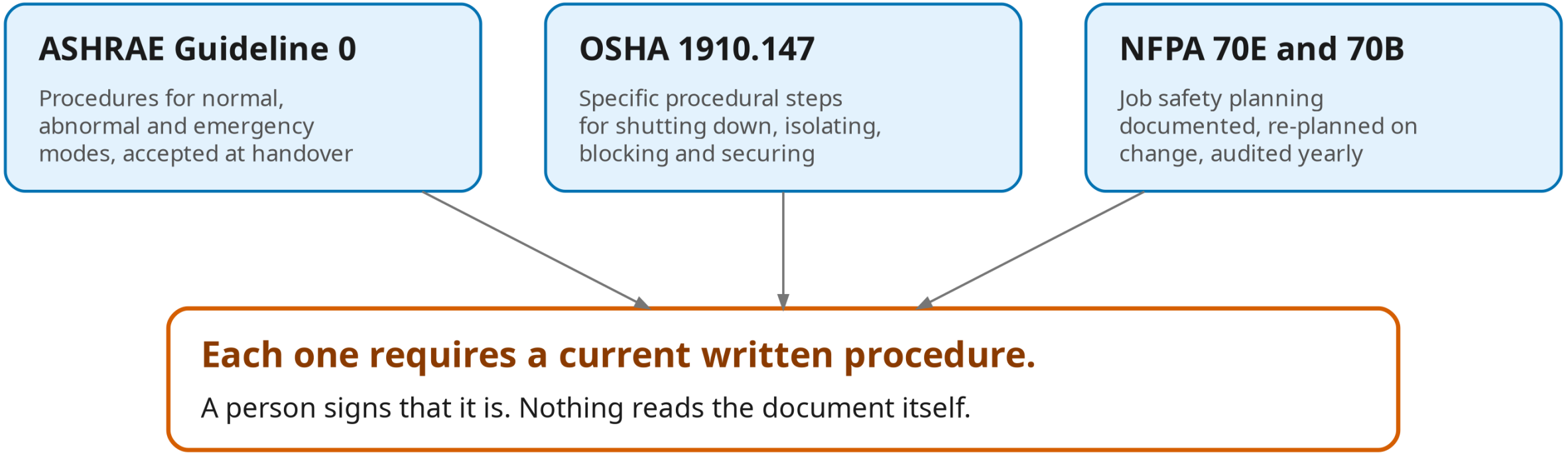
* OCP Digital Twin Initiative, Digital Twins for Data Centers, July 2026.

Procedures Get Outdated. Nothing Says Which Ones.



- When the hardware revises, the affected procedures can be searched for, never enumerated.
- Uptime Institute: the leading human factor in outages is procedure not followed, most often because it no longer matches the site.

The Only Check Is a Human Signature



- Compliance today is a signature. Nothing but a person ever inspects the document itself.*
- Checking one procedure against one rule is a person's job. Checking hundreds of them against every rule, and against every change made since they were written, is not.

* ASHRAE Guideline 0; OSHA 29 CFR 1910.147; NFPA 70E-2024; NFPA 70B.

Model the Work, Not Just the Facility

1. The Case for Brick

Why extend it, not replace it

2. The Schema

The vocabulary for the operational layer

3. The Extraction

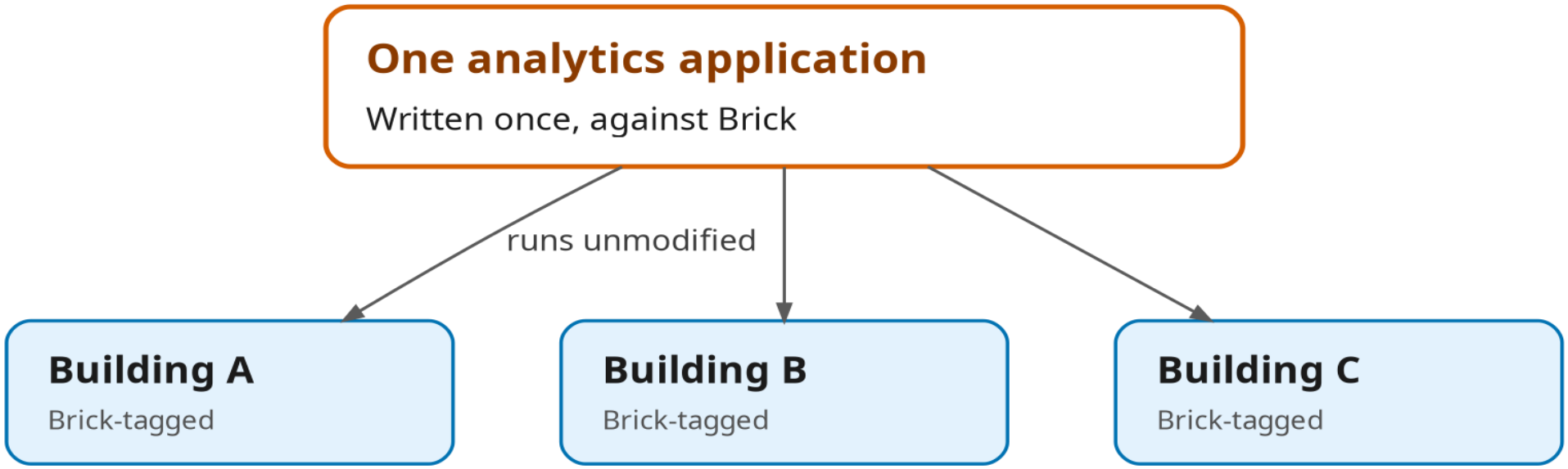
OCP Project Deschutes, step by step

4. Made Verifiable

Linked, traced, and checked automatically

- Procedures, components, interlocks, isolation points, and evidence become one graph that software can read.
- A collaboration between Visum AI and Lawrence Berkeley National Laboratory, aligned with ASHRAE 223.

Brick's Proven Cross-Site Portability



- The original Brick paper: six buildings, 17,700 data points, eight analytics applications, no customization.
- Extending Brick rather than replacing it means the operational layer inherits that reach, and converges with ASHRAE 223 rather than competing with it.

Liquid Cooling Arrived After Brick Was Written

Brick v1.4.4 covers

- Chiller
- VAV Box
- Sensor
- Setpoint

What liquid cooling adds

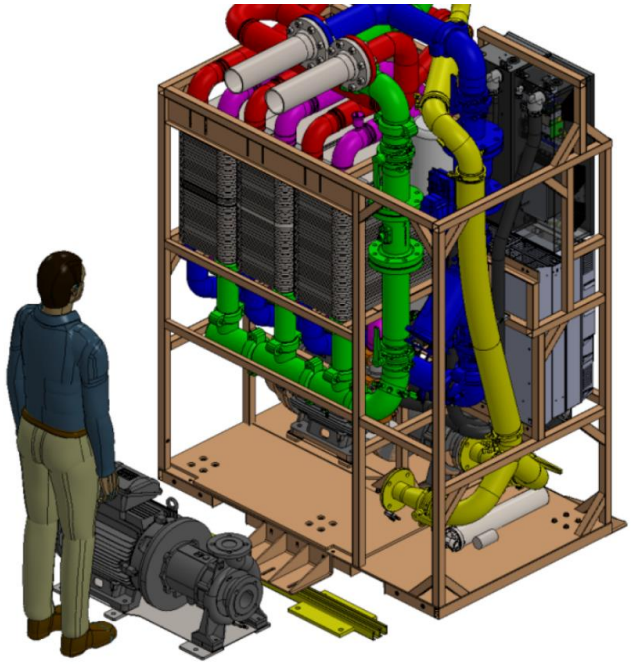
- Coolant Distribution Unit
- Rear-Door Heat Exchanger
- Cold Plate
- Liquid Cooling Manifold

The operational layer, across both columns: what the operator does, which isolation points govern access, what the safety standard requires

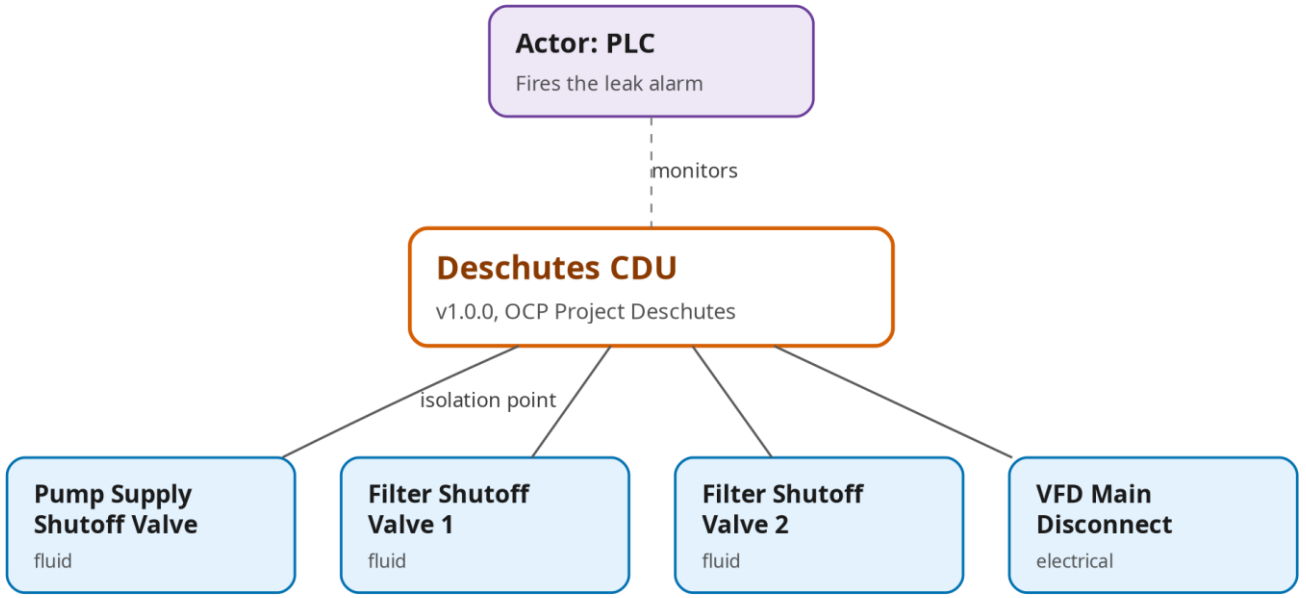
One Name for Everything a Procedure Touches

Procedure & Action	Procedure	Sequence	Action
State & Trigger	OperatingMode	Event	Interlock
Physical Isolation	IsolationPoint	LockoutTagoutStep	
Lifecycle & Evidence	FailureMode	CommissioningStep	MaintenanceTask
Accountability	Actor		

Every Requirement Cites Its Line in the Spec

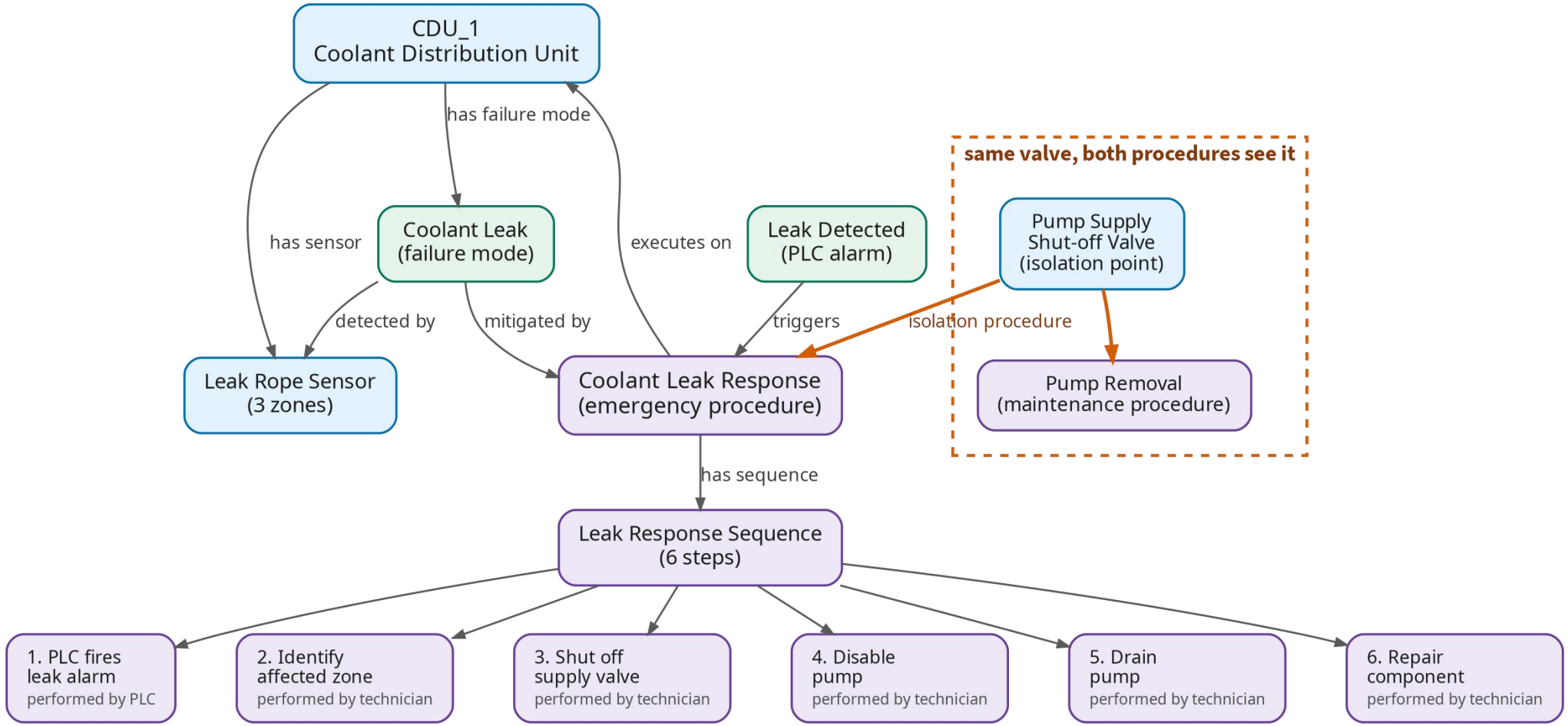


OCP Project Deschutes CDU, v1.0.0 spec (Google)

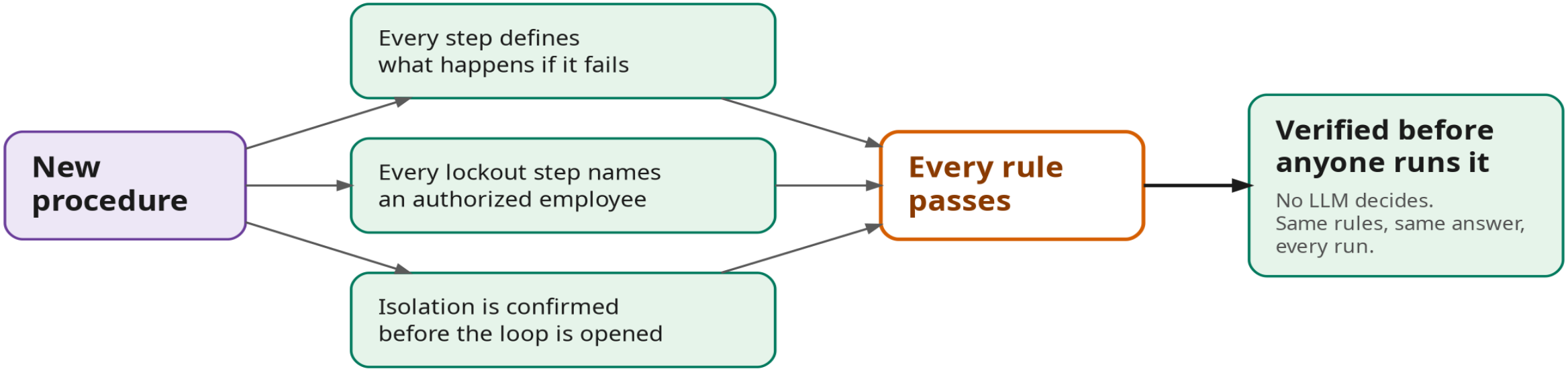


- Each isolation point cites exactly where in the spec its requirement comes from.

Complete Dependency Enumeration

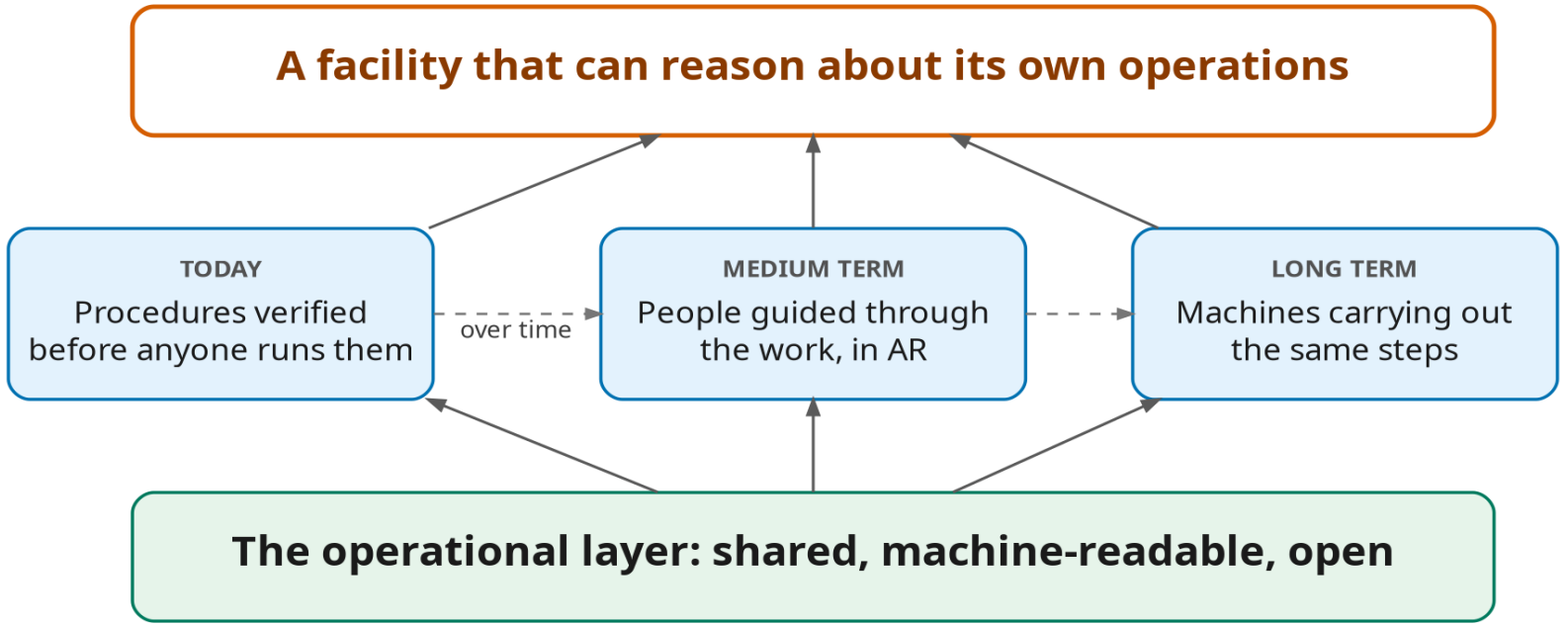


Deterministic Safety and Correctness Verification



- No LLM decides whether a rule passes. The rules run against the graph, so the same procedure gets the same answer on every run.
- Add a rule and every procedure is rechecked against it. Every procedure meets the same bar, and the bar rises for everyone at once.

The Model Changes How Facilities Run



- Every safety rule the industry agrees on, written once, enforced in every facility that adopts it.
- Procedures are the first thing built on this layer. Guidance in AR for the people doing the work is the medium term, and machines carrying out the same steps is the long term.

Build the Vocabulary With Us

- The vocabulary and the complete worked extraction are published at github.com/Visum-ai/aif-ops.
- Model one of your own procedures against it, and tell us what it could not represent.
- Help us make the vocabulary how the industry writes operations down, for every facility, not just cooling.
- nauman@visum.ai or mpritoni@lbl.gov

Open Discussion

